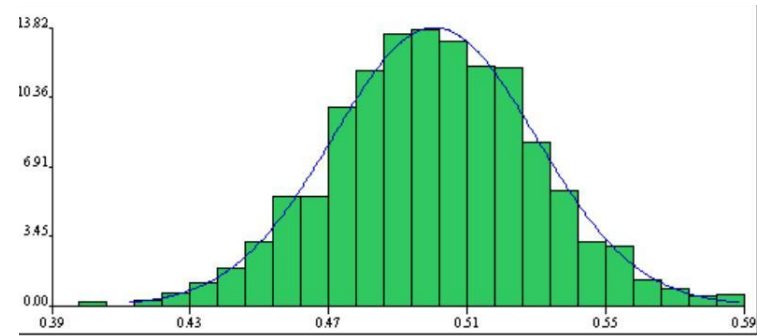
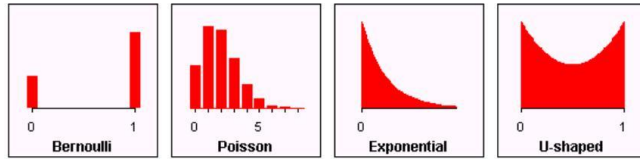


Data Science

CSE558

Overview

- Coupon Collector Problem
- Central Limit Theorem
- Statistical Inference



Parameter Estimation

Medical science in the west is superior than that in India's.
Let 31% of sample agrees to it.

- Empirical Mean: Mean from the sample.
0.31
- Confidence level: Region containing (most believable) parameter
95% confidence level $\Rightarrow$ 5% level of significance = 1.96
- *margin of error*: Ignored region. Here its $1.96 \times \sigma(\text{sample})$
Confidence interval: $0.31 \pm 1.96 \times 0.01 = (0.29, 0.33)$

In Practice: 95% confidence interval or 5% margin of error / level of significance.

In Practice: at least 15 success and 15 failure samples.

Overview

- Hypothesis Testing

Hypothesis Testing

Assumption: Data generation, distribution etc.

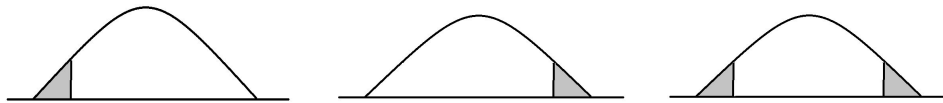
Hypothesis:

- Null Hypothesis (H_0): Status quo, i.e., declared true, until it is tested false.
- Alternate Hypothesis (H_1): Research hypothesis, i.e., range of situation that is to be diagnosed.

Statistics (T): A function that measures distance between empirical estimate and null hypothesis value.

P-value: Probability that measures whether test statistics inclined towards H_0 or H_1

Critical Region: One sided & Two sided.



Reject H_0 based on Critical Value

Hypothesis Testing

- Null Hypothesis (H_0): Status quo, i.e., declared true, until it is tested false.
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Statistics (T): A function that measures distance between empirical estimate and null hypothesis value.

P-value: Probability that measures whether test statistics inclined towards H_0 or H_1

Critical Region: One sided & Two sided.

Level of significance: Based on a probability α .

Error Types

- Type - I: Reject H_0 even when it is true.
- Type - II: Do not reject H_0 when it is false.

Decision\ Actual	H_0 True	H_0 False
H_0 Not Rejected	Correct Decision	Type - II
H_0 Rejected	Type - I	Correct Decision

Error Type

Court: *Not guilty until proven!*

H_0 : Defendant did not commit the crime

H_1 : Defendant committed the crime

Type - I:

Convicted the person, in reality did not commit the crime.

False positive, since rejecting H_0 when H_0 is true.

Decision\ Actual	Didn't Commit Crime	Committed Crime
Not Guilty	$1-\alpha$	β Type - II
Guilty	α Type - I	$1-\beta$

Never accept null hypothesis!

Type - II:

Acquitting the person, in reality did commit the crime.

False negative, since not rejecting H_0 when H_0 is false.

Example

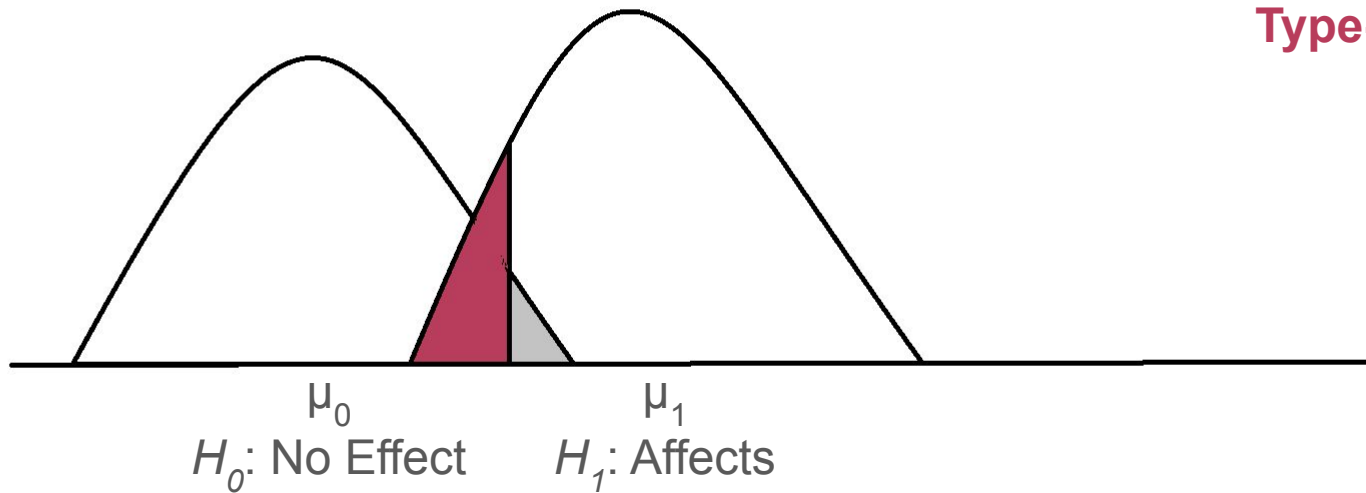
Does wearing masks from October to February in NCR improves lung function?

H_0 : Lung function remains same, with or without masks.

Type-I error: α

H_1 : Lung function improves after using masks.

Type-II error: β



z-test for Population Mean

- Univariable test
- Investigate sample mean vs population mean
- Standard deviation of population is known

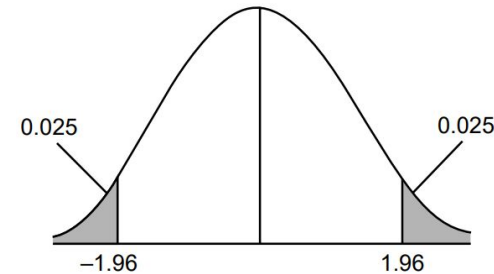
From a population of assumed mean μ_0 and known variance σ^2 , a random sample of size n is taken.

$$H_0: \mu = \mu_0; H_1: \mu \neq \mu_0;$$

Level of significance: $\alpha = 0.05$

Let sample mean is $\bar{x}$. Then the test statistics,
$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

What are the parameters of **Z**?

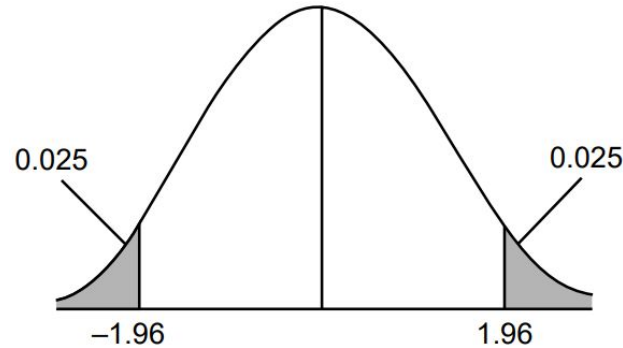


Example (z-test)

From a population of assumed mean $\mu_0 = 4.0$ and variance $\sigma^2 = 1$, a random sample of size $n = 9$ is taken.

$H_0: \mu = \mu_0; H_1: \mu \neq \mu_0;$

Level of significance: $\alpha = 0.05$



Let sample mean is $\bar{x} = 4.6$. Then the test statistics $Z = 1.8$.

Conclusion: Do not reject Null hypothesis.

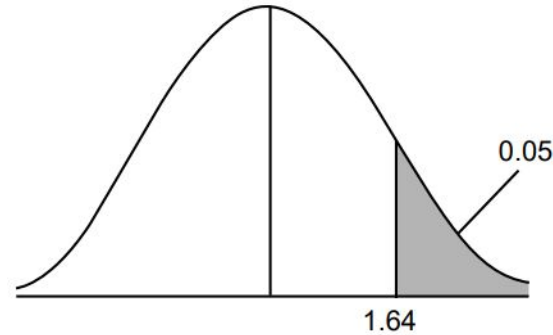
What does it imply?

Example (z-test)

From a population of assumed mean $\mu_0 = 4.0$ and variance $\sigma^2 = 1$, a random sample of size $n = 9$ is taken.

$$H_0: \mu \leq \mu_0; H_1: \mu > \mu_0;$$

Level of significance: $\alpha = 0.05$



Let sample mean is $\bar{x} = 4.6$. Then the test statistics $Z = 1.8$.

Conclusion: Reject Null hypothesis.

What does it imply?

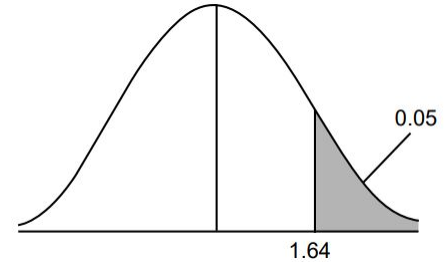
Statistical vs Practical Significance

A chip making claims at least 90% of their chip works right after production.

H_0 : Reject the claim; H_1 : Accept the claim.

You collected a random sample whose mean is 89.998% and found $z = 1.63$

Say $\alpha = 0.05$, will you reject or do not reject the claim?



Notice your sample mean was only 0.002% off from the population mean.

It implies that the sample mean is statistically significant for not to reject H_0 , even though its practical significance seems less.

z-test limitation

- Requires knowledge of standard deviation.
- Null hypothesis is true when $z \sim N(0,1)$.
- Use when sample size is large (say > 30)
- Accurate when distribution normal, else the test approximates..

z-test for a Proportion (Binomial Distribution)

- Univariable test
- Investigate proportion (say p for success) in sample vs population.
- Used when number of samples are large (say $n > 30$).
- Used to approximate normal distribution with binomial.

From an experiment with assumed probability of success being p_0 , a random sample of size n is taken.

Equivalence with z-test?

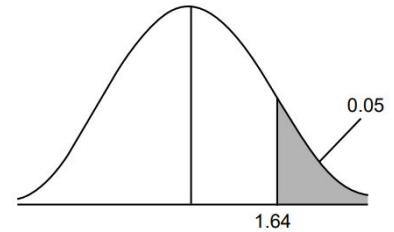
$H_0: p = p_0; H_1: p \neq p_0;$

Level of significance: $\alpha = 0.05$

Let p be the probability of success in the sample. Then the test statistics, $Z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

z-test for a Proportion (Binomial Distribution)

You believe at most 35% percent of people, wear powered glasses or contact lenses. You collected random samples of 100 people and found 60 of them not using powered glass or contact lens. With a 5% level of significance, will you reject your belief or not?



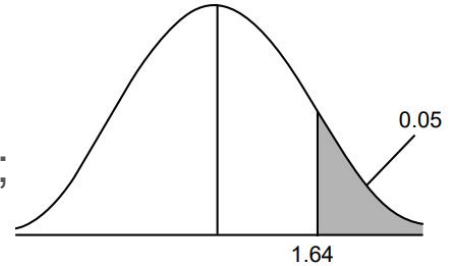
z-test for a Proportion (Binomial Distribution)

The probability of success $p_0 = 0.35$, sample size, $n=100$.

$H_0: p \leq p_0$ (at most 35% people have vision issues);

$H_1: p > p_0$; (at least 35% people does not have vision issues);

Level of significance: $\alpha = 0.05$



Let $p = 0.4$ is success probability of the sample. Then, test statistics $z = 1.05$.

Conclusion: Do not reject the null hypothesis.

Limitations: Use when sample size is significantly high (say $n > 30$).

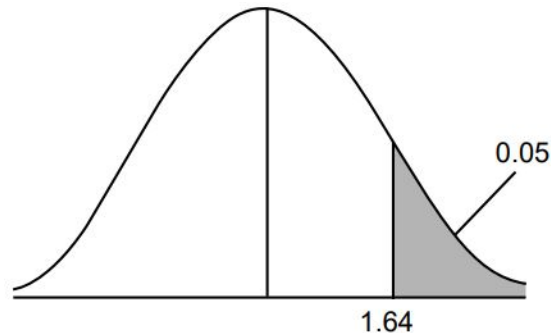
Example

Let your friend claims on an average she can correctly guess the suit of a randomly picked card with more than 0.25 probability.

$H_0: \mu \leq 0.25$; Guesses are random.

$H_1: \mu > 0.25$; Friend has supernatural powers.

Level of significance: $\alpha = 0.05$



Let sample mean is $\bar{x} = 0.28$, i.e., 28/100 guesses are correct.

What is your conclusion?

What if 44/100 were correct guess?

Question

1. Design a left tailed critical region test.
2. Design a two tailed critical region test.

Left tailed test: $H_1: \mu < \mu_0$

Right tailed test: $H_1: \mu > \mu_0$

Two tailed test: $H_1: \mu \neq \mu_0$

In general, two tailed tests are also robust towards distributions which are not normal.

t-test for Population Mean

- Univariable test, useful even when sample size is small.
- Investigate sample mean vs population mean
- Variance of population is unknown (z-test, for known variance)

From a population of assumed mean μ_0 , a random sample of size n is taken.

$$H_0: \mu = \mu_0; \quad H_1: \mu \neq \mu_0;$$

Level of significance: $\alpha = 0.05$

Let sample mean is $\bar{x}$ and $s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n - 1}$. Then the test statistics, $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Degrees of freedom = $n-1$

Example (t-test)

From a population of assumed mean $\mu_0 = 4.0$, a random sample of size $n=9$ is taken.

$$H_0: \mu = \mu_0; H_1: \mu \neq \mu_0;$$

Level of significance: $\alpha = 0.05$

Let sample mean is $\bar{x} = 4.9$ and $s^2 = 1.0$.

Then the test statistics $t = 2.7$.

Degrees of freedom = 8

Conclusion: Reject Null hypothesis.

Limitations: Accurate when distribution normal, else the test approximates.

ν	Level of significance α				
	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169

Example

You are teaching AI for the first time to a class strength of 5000. You conducted a test and you that an average student should pass with good marks (say greater than 50/100).

Your 10 random samples reports following
16,28,62,30,36,91,43,31,21,42

With a 10% level of significance what is
your conclusion?

ν	Level of significance α				
	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
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